SPRING 2026

CSC 220 Data Structures & Program Design - Syllabus

Credits: 4
Instructor: Prof. Marcus Oyelaran
Meeting times: Tue/Thu 13:00-14:15
Office hours: Wed 14:00-16:00 or by appointment
Prerequisites: As listed in the catalogue
Format: In person

Course Description

Data structures and program design: implementation and analysis of fundamental structures and the algorithms that use them.

Learning Outcomes

- Implement lists, stacks, queues, trees and hash tables
- analyse running time using Big-O notation
- design object-oriented programs

Topics Covered

- Arrays, linked lists, stacks and queues
- Trees and binary search trees
- Hash tables and sets
- Sorting and searching algorithms
- Algorithm efficiency and Big-O analysis

Weekly Schedule

Week 1-3:	Arrays, linked lists, stacks and queues
Week 4-6:	Trees and binary search trees
Week 7-9:	Hash tables and sets
Week 10-12:	Sorting and searching algorithms
Week 13-15:	Algorithm efficiency and Big-O analysis

Grading Policy

Assignments and quizzes	**30%**
Midterm exam	**30%**
Final exam	**40%**

Academic Integrity

Students are expected to submit original work and to cite sources. Violations are handled under the university academic honesty policy.